∴ The condition that for each row, the leading entry must be in a column to the right of the columns of the leading entries of all rows above it is True.

3. Row 2 Column 1 = Row 3 Column 1 = 0
Row 3 Column 2 = 0

∴ The condition that for each column having a leading entry, all elements below the leading entry position must be 0 is True.

4. All rows have 1 ~~as~~ the leading entry. ∴ The condition is satisfied

5. Row 1 Column 3 = Row 2 Column 3 = 0
Row 1 Column 2 = 0

∴ The condition that for each column having a leading entry, all elements above the leading entry position must be all 0 is True.

∴ The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is in reduced row echelon form.

There is a pivot position in every row of the matrix A. ∴ The columns of A span R^3. □

30) Construct a 3x3 matrix, not in echelon form, whose columns do not span R^3. Show that the matrix you construct has the desired property.

$\boxed{\text{Claim:}}$ The matrix $B = \begin{bmatrix} 3 & 1 & 2 \\ 2 & 5 & 6 \\ 5 & 6 & 8 \end{bmatrix}$ is not in echelon form and its columns do not span R^3.

$\boxed{\text{Proof:}}$ Row 1 Leading entry = 3 in Column 1
Row 2 Leading entry = 2 in Column 1
Row 3 Leading entry = 5 in Column 1

Since in the echelon form of a matrix, each row has its leading element in a column to the right of the columns of the leading elements of all rows above it, the matrix B is not in echelon form.

From Theorem 4, the columns of B span R^3 if and only if B has a pivot position in every row. ∴ The columns of B do not span R^3 if and only if there exists atleast one row in B which do not have a pivot position.